Deriving the Standard Model Lagrangian

A Step-by-Step Construction from First Principles

Study Notes for Elementary Particle Physics

The Game Plan

The Standard Model Lagrangian is built in layers:

1. Start with free fermion fields (Dirac Lagrangian)
2. Identify the gauge symmetry group: $SU(3)_C \times SU(2)_L \times U(1)_Y$
3. Demand *local* gauge invariance $\Rightarrow$ introduces gauge bosons via covariant derivatives
4. Add gauge field kinetic terms (field-strength tensors)
5. Problem: gauge invariance forbids mass terms for gauge bosons *and* fermions
6. Introduce the Higgs field with spontaneous symmetry breaking $\Rightarrow$ generates all masses

The final answer is:

$$\boxed{\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}} \tag{1}$$

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1 Free Fermions: Where It All Starts

1.1 The Dirac Lagrangian

A free spin-$\frac{1}{2}$ particle of mass m is described by a 4-component spinor ψ obeying the Dirac equation, which comes from the Lagrangian:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi \tag{2}$$

where $\bar{\psi} = \psi^\dagger \gamma^0$.

1.2 Chirality: left- and right-handed fields

Define the projection operators $P_L = \frac{1}{2}(1-\gamma^5)$ and $P_R = \frac{1}{2}(1+\gamma^5)$. Any spinor splits as:

$$\psi = \psi_L + \psi_R, \qquad \psi_L = P_L\psi, \qquad \psi_R = P_R\psi \tag{3}$$

Why chirality matters

The weak interaction treats ψ_L and ψ_R **differently**. Left-handed fermions live in $SU(2)_L$ doublets; right-handed fermions are $SU(2)_L$ singlets. This is parity violation, built into the structure of the gauge group.

The kinetic term respects chirality: $\bar{\psi} i\gamma^\mu \partial_\mu \psi = \bar{\psi}_L i\gamma^\mu \partial_\mu \psi_L + \bar{\psi}_R i\gamma^\mu \partial_\mu \psi_R$.
But the mass term **mixes** chiralities:

$$-m\bar{\psi}\psi = -m(\bar{\psi}_R\psi_L + \bar{\psi}_L\psi_R) \tag{4}$$

This will be a problem later — mass terms break gauge invariance.

1.3 The fermion content of the Standard Model

For one generation (electron family), arranged by their gauge quantum numbers:

Field	Components	$SU(3)_C$	$SU(2)_L$	$U(1)_Y$
$L = \begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix}$	LH lepton doublet	**1**	**2**	-1
e_R	RH electron singlet	**1**	**1**	-2
$Q = \begin{pmatrix} u_L \\ d_L \end{pmatrix}$	LH quark doublet	**3**	**2**	$+\frac{1}{3}$
u_R	RH up singlet	**3**	**1**	$+\frac{4}{3}$
d_R	RH down singlet	**3**	**1**	$-\frac{2}{3}$

The hypercharge assignments satisfy the **Gell-Mann–Nishijima relation**: $Q = I_W^{(3)} + \frac{Y}{2}$.

Why these quantum numbers?

They aren't derived — they're *inputs* from experiment. The fact that they lead to a consistent, anomaly-free theory is nontrivial and remarkable.

2 The Gauge Principle: How Interactions Arise

2.1 The idea

The Local Gauge Principle

Global symmetry: $\psi \to e^{i\alpha}\psi$ with α constant. The Dirac Lagrangian is invariant.
Local symmetry: $\psi \to e^{i\alpha(x)}\psi$ with α depending on spacetime. The kinetic term is **not** invariant because ∂_μ acts on $\alpha(x)$.
Fix: Replace $\partial_\mu \to D_\mu = \partial_\mu + igA_\mu$. The new field A_μ transforms to cancel the unwanted $\partial_\mu\alpha$ term. This A_μ *is* the gauge boson.
The demand for local symmetry **forces** the existence of interactions.

2.2 $U(1)$ example: QED

Demand invariance under $\psi \to e^{iq\alpha(x)}\psi$. Replace $\partial_\mu \to D_\mu = \partial_\mu + iqA_\mu$. The gauge field transforms as $A_\mu \to A_\mu - \partial_\mu\alpha$.

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \tag{5}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The interaction $-q\bar{\psi}\gamma^\mu A_\mu\psi$ emerges automatically. This is the photon coupling to charged fermions.

2.3 $SU(2)_L$: the weak interaction

The LH fermion doublet $L = \begin{pmatrix}\nu_L \\ e_L\end{pmatrix}$ transforms under $SU(2)_L$:

$$L \to e^{ig_W\vec{\alpha}(x)\cdot\vec{T}}\,L, \qquad \vec{T} = \frac{\vec{\sigma}}{2} \tag{6}$$

where $\vec{\sigma}$ are the Pauli matrices (the three generators of $SU(2)$). This requires **three** gauge fields $\vec{W}_\mu = (W^1_\mu, W^2_\mu, W^3_\mu)$:

$$D_\mu L = \left(\partial_\mu + ig_W\vec{T}\cdot\vec{W}_\mu\right)L \tag{7}$$

RH singlets don't transform under $SU(2)_L$:

$$D_\mu e_R = \partial_\mu e_R \qquad \text{(no } W \text{ coupling for RH fields)} \tag{8}$$

This is parity violation: only LH particles feel the weak force.

2.4 $U(1)_Y$: weak hypercharge

A separate $U(1)$ symmetry with coupling g' and gauge field B_μ:

$$\psi \to e^{ig'\frac{Y}{2}\alpha(x)}\psi, \qquad D_\mu \supset ig'\frac{Y}{2}B_\mu \tag{9}$$

where Y is the weak hypercharge of the field (from the table above). **Both** LH and RH fields couple to B_μ, but with different hypercharges.

2.5 $SU(3)_C$: QCD

Quarks carry color charge and transform under $SU(3)_C$:

$$q \to e^{ig_s\alpha^a(x)T^a}\,q, \qquad a = 1,\ldots,8 \tag{10}$$

where $T^a = \frac{\lambda^a}{2}$ are the eight Gell-Mann matrices. This requires **eight** gluon fields G^a_μ:

$$D_\mu q = \left(\partial_\mu + ig_s T^a G^a_\mu\right)q \tag{11}$$

Leptons are color singlets and don't couple to gluons.

2.6 The full covariant derivative

Combining all three gauge groups, the covariant derivative acting on, say, the LH quark doublet Q is:

$$\boxed{D_\mu = \partial_\mu + ig_s T^a G^a_\mu + ig_W T^i W^i_\mu + ig' \frac{Y}{2} B_\mu} \tag{12}$$

Each field only "sees" the gauge bosons it's charged under. For example, e_R has no $SU(3)_C$ or $SU(2)_L$ charge, so only the B_μ term survives.

3 Gauge Field Kinetic Terms

Each gauge field needs its own kinetic energy term, built from the **field-strength tensor**.

3.1 Abelian: $U(1)_Y$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \tag{13}$$

$$\mathcal{L}_{U(1)} = -\frac{1}{4} B_{\mu\nu} B^{\mu\nu} \tag{14}$$

3.2 Non-Abelian: $SU(2)_L$

Because the generators don't commute ($[\sigma^i, \sigma^j] = 2i\epsilon^{ijk}\sigma^k$), the field-strength tensor picks up an extra term:

$$W^i_{\mu\nu} = \partial_\mu W^i_\nu - \partial_\nu W^i_\mu - g_W \epsilon^{ijk} W^j_\mu W^k_\nu \tag{15}$$

$$\mathcal{L}_{SU(2)} = -\frac{1}{4} W^i_{\mu\nu} W^{i\mu\nu} \tag{16}$$

Why non-Abelian matters

The extra $\epsilon^{ijk} W^j W^k$ term means the W bosons **interact with each other**. Photons don't self-interact (QED is Abelian). Gluons do (QCD is non-Abelian). The weak bosons also self-interact. This is a direct consequence of the non-commuting group structure.

3.3 Non-Abelian: $SU(3)_C$

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu - g_s f^{abc} G^b_\mu G^c_\nu \tag{17}$$

where f^{abc} are the $SU(3)$ structure constants ($[T^a, T^b] = i f^{abc} T^c$).

$$\mathcal{L}_{SU(3)} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} \tag{18}$$

Complete Gauge Sector

$$\boxed{\mathcal{L}_{\text{gauge}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} - \frac{1}{4} W^i_{\mu\nu} W^{i\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu}} \tag{19}$$

12 gauge bosons total: 8 gluons + 3 weak + 1 hypercharge.

4 The Fermion Sector

Now combine the free fermion Lagrangians with the covariant derivatives. For one generation:

Fermion Kinetic + Gauge Interaction Terms

$$\boxed{\mathcal{L}_{\text{fermion}} = \bar{L}\, i\gamma^\mu D_\mu\, L + \bar{e}_R\, i\gamma^\mu D_\mu\, e_R + \bar{Q}\, i\gamma^\mu D_\mu\, Q + \bar{u}_R\, i\gamma^\mu D_\mu\, u_R + \bar{d}_R\, i\gamma^\mu D_\mu\, d_R} \tag{20}$$

where D_μ is the appropriate covariant derivative for each field (determined by its gauge quantum numbers).

Notice: **no mass terms**. We cannot write $-m_e\bar{e}e = -m_e(\bar{e}_R e_L + \bar{e}_L e_R)$ because e_L lives in an $SU(2)$ doublet but e_R is a singlet — the mass term would break gauge invariance.

This is the **mass problem**: local gauge invariance forbids masses for both gauge bosons and fermions. Something else must generate them.

5 The Higgs Mechanism: Generating Mass

5.1 The Higgs field

Introduce a complex scalar $SU(2)_L$ doublet with hypercharge $Y = +1$:

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix} \tag{21}$$

Its Lagrangian, respecting the full gauge symmetry:

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \phi)^\dagger (D^\mu \phi) - V(\phi) \tag{22}$$

with the **Mexican hat potential**:

$$V(\phi) = \mu^2(\phi^\dagger \phi) + \lambda(\phi^\dagger \phi)^2, \qquad \mu^2 < 0, \quad \lambda > 0 \tag{23}$$

Why $\mu^2 < 0$?

A normal mass term $+m^2|\phi|^2$ has its minimum at $\phi = 0$. With $\mu^2 < 0$, the minimum is at $|\phi| \neq 0$ — the field has a nonzero **vacuum expectation value** (VEV). This is what triggers spontaneous symmetry breaking.

5.2 Spontaneous symmetry breaking

The minimum of V occurs at $\phi^\dagger \phi = v^2/2$ where $v = \sqrt{-\mu^2/\lambda}$. We choose the vacuum:

$$\langle \phi \rangle_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} \tag{24}$$

This particular choice breaks $SU(2)_L \times U(1)_Y$ but preserves a $U(1)_{\text{em}}$ subgroup (electromagnetism survives). The preserved symmetry corresponds to $Q = I_W^{(3)} + Y/2$, which is the electric charge operator. The photon stays massless; the $W^\pm$ and Z get massive.

5.3 Expanding around the vacuum

In **unitary gauge**, the Higgs doublet is:

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} \tag{25}$$

where $h(x)$ is the physical Higgs boson (the excitation around the VEV). Three of the four original degrees of freedom become the longitudinal modes of $W^\pm$ and Z ("eaten" Goldstone bosons).

5.4 Gauge boson masses from $(D_\mu \phi)^\dagger (D^\mu \phi)$

The covariant derivative acting on the Higgs doublet is:

$$D_\mu \phi = \left(\partial_\mu + i g_W \frac{\vec{\sigma}}{2} \cdot \vec{W}_\mu + i g' \frac{1}{2} B_\mu \right) \phi \tag{26}$$

Substituting $\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix}$ and expanding $(D_\mu \phi)^\dagger (D^\mu \phi)$, the terms quadratic in gauge fields (setting $h = 0$ to extract masses) give:

$$\frac{v^2}{8} \left[g_W^2 (W_\mu^1)^2 + g_W^2 (W_\mu^2)^2 + (g_W W_\mu^3 - g' B_\mu)^2 \right] \tag{27}$$

5.5 Physical gauge bosons

Define the physical fields:

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2) \qquad m_W = \frac{1}{2}g_W v \tag{28}$$

$$Z_\mu = \frac{g_W W_\mu^3 - g' B_\mu}{\sqrt{g_W^2 + g'^2}} \qquad m_Z = \frac{v}{2}\sqrt{g_W^2 + g'^2} = \frac{m_W}{\cos\theta_W} \tag{29}$$

$$A_\mu = \frac{g' W_\mu^3 + g_W B_\mu}{\sqrt{g_W^2 + g'^2}} \qquad m_A = 0 \tag{30}$$

where the **weak mixing angle** (Weinberg angle) is:

$$\cos\theta_W = \frac{g_W}{\sqrt{g_W^2 + g'^2}}, \qquad \sin\theta_W = \frac{g'}{\sqrt{g_W^2 + g'^2}} \tag{31}$$

Key Relation

$$\boxed{e = g_W \sin\theta_W = g' \cos\theta_W} \tag{32}$$

The electromagnetic coupling e is *derived* from the weak couplings. Electromagnetism and the weak force are unified.

5.6 Higgs boson mass

From expanding $V(\phi)$ around the VEV:

$$m_h = \sqrt{2\lambda}\, v = \sqrt{-2\mu^2} \tag{33}$$

With $v = 246$ GeV (from G_F) and $m_h \approx 125$ GeV (measured at LHC), we get $\lambda \approx 0.13$.

6 Fermion Masses: Yukawa Couplings

6.1 The problem, restated

We cannot write $-m_e \bar{e} e$ directly because $\bar{e}_L$ is part of an $SU(2)$ doublet and e_R is a singlet. This product isn't gauge invariant.

6.2 The solution

Construct a gauge-invariant combination using the Higgs doublet:

$$\mathcal{L}_{\text{Yukawa}} = -y_e \, \bar{L} \, \phi \, e_R + \text{h.c.} \tag{34}$$

Check: $\bar{L}$ is a $\mathbf{2}^*$ of $SU(2)$, ϕ is a $\mathbf{2}$, e_R is a $\mathbf{1}$. The product $\bar{L}\phi$ contracts the $SU(2)$ indices to a singlet. Hypercharges: $\bar{L}$ has $Y = +1$, ϕ has $Y = +1$, e_R has $Y = -2$. Total: $+1 + 1 - 2 = 0$. Gauge invariant.

After symmetry breaking, substitute $\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix}$:

$$\mathcal{L}_{\text{Yukawa}} = -\frac{y_e}{\sqrt{2}} (\bar{\nu}_{eL} \;\; \bar{e}_L) \begin{pmatrix} 0 \\ v + h \end{pmatrix} e_R + \text{h.c.} \tag{35}$$

$$= -\frac{y_e v}{\sqrt{2}} \bar{e}_L e_R - \frac{y_e}{\sqrt{2}} \bar{e}_L e_R \, h + \text{h.c.} \tag{36}$$

The first term is a **mass term**: $m_e = \frac{y_e v}{\sqrt{2}}$.
The second term is the **Higgs–electron coupling**: strength $\propto m_e / v$.

Masses proportional to Higgs coupling

The Yukawa coupling y_e is a free parameter. The heavier a fermion, the more strongly it couples to the Higgs. The top quark ($m_t \approx 173$ GeV, $y_t \approx 1$) couples almost as strongly as possible; the electron ($m_e = 0.511$ MeV, $y_e \approx 3 \times 10^{-6}$) couples extremely weakly.

6.3 Up-type quarks: need $\tilde{\phi}$

The Higgs VEV is in the *lower* component, so $\bar{L}\phi R$ only generates masses for the *lower* member of each doublet (e, d, s, b). For up-type quarks, use the charge-conjugate Higgs:

$$\tilde{\phi} = i\sigma^2 \phi^* = \begin{pmatrix} \phi^{0*} \\ -\phi^- \end{pmatrix} \xrightarrow{\text{VEV}} \frac{1}{\sqrt{2}} \begin{pmatrix} v + h \\ 0 \end{pmatrix} \tag{37}$$

Then: $\mathcal{L} \supset -y_u \, \bar{Q} \, \tilde{\phi} \, u_R + \text{h.c.}$ gives $m_u = y_u v / \sqrt{2}$.

Complete Yukawa Sector (one generation)

$$\boxed{\mathcal{L}_{\text{Yukawa}} = -y_e \, \bar{L} \, \phi \, e_R - y_d \, \bar{Q} \, \phi \, d_R - y_u \, \bar{Q} \, \tilde{\phi} \, u_R + \text{h.c.}} \tag{38}$$

With three generations, y_e, y_d, y_u become 3×3 matrices, and diagonalizing the quark mass matrices introduces the CKM mixing matrix.

7 The Complete Standard Model Lagrangian

The Standard Model Lagrangian

$$\boxed{\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}} \tag{39}$$

Gauge boson kinetic terms:

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} - \frac{1}{4} W^i_{\mu\nu} W^{i\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} \tag{40}$$

Fermion kinetic + interaction terms (summed over 3 generations):

$$\mathcal{L}_{\text{fermion}} = \sum_{\text{gen}} \left[\bar{L}\, i\gamma^\mu D_\mu L + \bar{e}_R\, i\gamma^\mu D_\mu\, e_R + \bar{Q}\, i\gamma^\mu D_\mu Q + \bar{u}_R\, i\gamma^\mu D_\mu\, u_R + \bar{d}_R\, i\gamma^\mu D_\mu\, d_R \right] \tag{41}$$

Higgs sector:

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \phi)^\dagger (D^\mu \phi) - \mu^2 \phi^\dagger \phi - \lambda (\phi^\dagger \phi)^2 \tag{42}$$

Yukawa couplings (generation indices suppressed):

$$\mathcal{L}_{\text{Yukawa}} = -Y_e\, \bar{L}\, \phi\, e_R - Y_d\, \bar{Q}\, \phi\, d_R - Y_u\, \bar{Q}\, \tilde{\phi}\, u_R + \text{h.c.} \tag{43}$$

Counting the free parameters

Parameters	**Count**
Gauge couplings (g_s, g_W, g')	3
Higgs potential $(\mu^2, \lambda$ or equivalently $v, m_h)$	2
Yukawa matrices $\to$ 6 quark masses + 3 lepton masses	9
CKM matrix (3 angles + 1 phase)	4
QCD vacuum angle θ	1
Total	**19**

(With massive neutrinos, add 7–9 more parameters depending on whether they're Dirac or Majorana.)

8 Summary: What Each Piece Does

Term	Physics	Particles Involved
$-\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu}$	Gluon kinetic energy + gluon self-interactions	8 gluons (massless)
$-\frac{1}{4}W^i_{\mu\nu}W^{i\mu\nu}$	Weak boson kinetic energy + self-interactions	W^1, W^2, W^3 (before mixing)
$-\frac{1}{4}B_{\mu\nu}B^{\mu\nu}$	Hypercharge boson kinetic energy	B (before mixing)
$\bar{\psi}\, i\gamma^\mu D_\mu\, \psi$	Fermion kinetic energy + all gauge interactions	All fermion–boson vertices
$(D_\mu\phi)^\dagger(D^\mu\phi)$	Higgs kinetic energy + **gauge boson masses** (after SSB)	$W^\pm, Z$ masses; γ massless
$-V(\phi)$	Higgs potential $\to$ SSB + Higgs boson mass	Higgs self-interactions
$-Y_f\bar{L}\phi R$	Yukawa $\to$ **fermion masses** (after SSB)	Fermion–Higgs coupling

The Logic in One Paragraph

Start with massless fermions and demand local $SU(3)_C \times SU(2)_L \times U(1)_Y$ invariance. This *forces* the existence of 12 gauge bosons with specific couplings. But gauge invariance *forbids* all mass terms. Introduce a scalar doublet with a Mexican-hat potential that breaks $SU(2)_L \times U(1)_Y \to U(1)_{\text{em}}$. This gives masses to $W^\pm$ and Z while keeping γ and gluons massless. Yukawa couplings between the fermions and the Higgs generate all fermion masses. The result: everything we observe, from nuclear forces to radioactive decay to electromagnetism to the 125 GeV Higgs boson, emerges from a single Lagrangian with 19 free parameters. That's the Standard Model.

9 After Symmetry Breaking: The Interaction Vertices

Once we expand $\mathcal{L}_{\text{SM}}$ around the Higgs VEV and rewrite everything in terms of physical fields ($\gamma, W^{\pm}, Z, g$, massive fermions, h), we get every interaction vertex in particle physics. The key ones:

QED vertex

From $\bar{\psi} i\gamma^{\mu} D_{\mu} \psi$ with the A_{μ} piece: $\quad -eQ_f \bar{f}\gamma^{\mu} f A_{\mu}$

Electric charge Q_f times e. Couples photon to any charged fermion.

Weak charged current

From $\bar{L} i\gamma^{\mu} D_{\mu} L$ with the $W^{\pm}$ pieces: $\quad -\frac{g_W}{\sqrt{2}} \bar{\nu}_L \gamma^{\mu} e_L W^{+}_{\mu} + \text{h.c.}$

Only LH fermions. Changes flavor ($\nu \leftrightarrow e$, $u \leftrightarrow d$).

Weak neutral current

From the Z_{μ} piece: $\quad -\frac{g_W}{\cos\theta_W} \bar{f}\gamma^{\mu}(g_V^f - g_A^f \gamma^5) f \, Z_{\mu}$

Couples to both LH and RH (but not equally — parity violation).

QCD vertex

From $\bar{q} i\gamma^{\mu} D_{\mu} q$ with the G^a_{μ} piece: $\quad -g_s \bar{q}\gamma^{\mu} T^a q \, G^a_{\mu}$

Color-changing vertex. Only quarks.

Higgs couplings

From Yukawa after SSB: $\quad -\frac{m_f}{v} \bar{f} f \, h$

Coupling strength $\propto$ mass. Top quark couples most strongly. Neutrinos (in minimal SM) don't couple at all.

From $(D_{\mu}\phi)^{\dagger}(D^{\mu}\phi)$: $\quad g_W m_W W^{+}_{\mu} W^{-\mu} h \quad$ and $\quad \frac{g_W m_Z}{\cos\theta_W} Z_{\mu} Z^{\mu} h$

Higgs couples to gauge bosons $\propto$ their mass^2.

What the SM does NOT explain

The Standard Model is extraordinarily successful but leaves open:

- Why 3 generations?
- Why these particular masses (19 free parameters)?
- Gravity is not included
- Dark matter and dark energy
- Neutrino masses (require BSM physics)
- The strong CP problem ($\theta \approx 0$ — why?)
- The hierarchy problem ($m_h \ll M_{\text{Planck}}$ — why?)
- Matter–antimatter asymmetry